Lab: Online Machine Learning with River and CapyMOA

Nov 9, 2026 · M2 Data Science - Institut Polytechnique de Paris - Dr Mariam BARRY - Dr Maurras TOGBE

In this lab you implement online machine learning methods with River and CapyMOA, compare them on real and synthetic streams, and analyze the results. The lab has two parts and ends with a written report.

Material and documentation

- **River:** documentation and GitHub repository. Install with `pip install river` (Python 3.11 or later).
- **CapyMOA:** documentation, installation guide, GitHub repository and paper (arXiv 2502.07432). Install with `pip install capymoa`; it needs a Java runtime, and PyTorch only for the deep-learning methods.

Evaluation protocol (for all steps)

- Use prequential (test-then-train) evaluation.
- Fix random seeds and use the same datasets and hyperparameters across methods and libraries, so comparisons are fair.
- Report predictive performance and also cost: processing time per instance (latency), total runtime and, when possible, memory.

Part 1: River

- **Step 1. Installation.** In a Jupyter notebook or a Python environment, install River by following the official guidelines (https://riverml.xyz).
- **Step 2. Regression methods and datasets.** Select 4 regression methods of different types (tree-based, ensemble-based, linear with adaptive learning rate, and one more of your choice), and 4 datasets: 3 real and 1 synthetic.
- **Step 3. Comparison.** Run the methods and compare them on the 4 datasets with several metrics (for example MAE, RMSE, R2) and with their running time.
- **Step 4. Drift detection.** Compare drift detectors on the same datasets and compare the results: number and position of detected drifts, detection delay when the drift position is known (synthetic data), and the effect of drift handling on model performance.

- **Step 5. Classification.** Repeat steps 2 to 4 with classification methods: 4 methods, 3 real datasets and 1 synthetic dataset, with metrics such as accuracy, F1-score and Kappa.
- **Step 6. Report.** Write a report and analyze the results of Part 1.

Suggestions for Part 1 (check availability in your River version)

Type	Regression methods	Classification methods
Tree-based	HoeffdingTreeRegressor, HoeffdingAdaptiveTreeRegressor	HoeffdingTreeClassifier, HoeffdingAdaptiveTreeClassifier
Ensemble-based	AdaptiveRandomForestRegressor, SRPRegressor	AdaptiveRandomForestClassifier, ADWINBaggingClassifier
Linear with adaptive learning	LinearRegression with an adaptive optimizer (AdaGrad, Adam), PARegressor	LogisticRegression with an adaptive optimizer
Other	KNNRegressor	GaussianNB, KNNClassifier

- **Drift detectors:** ADWIN, Page-Hinkley, KSWIN (regression); ADWIN, DDM, EDDM, HDDM (classification).
- **Real datasets:** for example TrumpApproval, ChickWeights, Bikes (regression); Elec2, Phishing, Insects (classification).
- **Synthetic streams:** for example Friedman and FriedmanDrift (regression); SEA, Agrawal, Hyperplane and ConceptDriftStream (classification).

Part 2: CapyMOA

- **Step 7. Installation.** Install CapyMOA using the official documentation (https://capymoa.org). CapyMOA relies on a Java runtime, so follow the installation guide carefully.
- **Step 8. Classification comparison.** Select the same 4 classification methods as in River (or their closest equivalents in CapyMOA). Compare the results with those obtained in River on accuracy, processing time and latency.
- **Step 9. Clustering.** Try several stream clustering methods on unsupervised datasets. Analyze the quality of the clusters and how they evolve over the stream.
- **Step 10. Bonus: AutoML.** Try the AutoML module of CapyMOA and compare the different approaches it offers.

Deliverables

1. **A report in PDF only**, between 5 and 10 pages, that includes:
    - Brief présentation of the methods
    - Experimental setup (datasets, methods, hyperparameters, metrics, hardware)
    - result tables and figures for each step
    - an analysis: accuracy versus cost trade-offs, behavior of the drift detectors, River versus CapyMOA
    - a conclusion on which method to use in which situation
2. **The notebook file(s) exported as HTML**, or a link to a repository containing the code (for example on GitHub).

Submission

- **Deadline:** 20 November 2026. 23:59 CET
- **Group size:** maximum 2 students per group. If you do the lab alone, you get a bonus.
- **By email** to mariam.sa.barry@gmail.com and maurras.togbe@isep.fr: the PDF report, plus the HTML notebook file or the link to the repository.
- **Email subject:** `Lab Online ML - Name 1 - Name 2 - River - CapyMOA`
- **On Moodle:** also upload the same files in the deposit box.